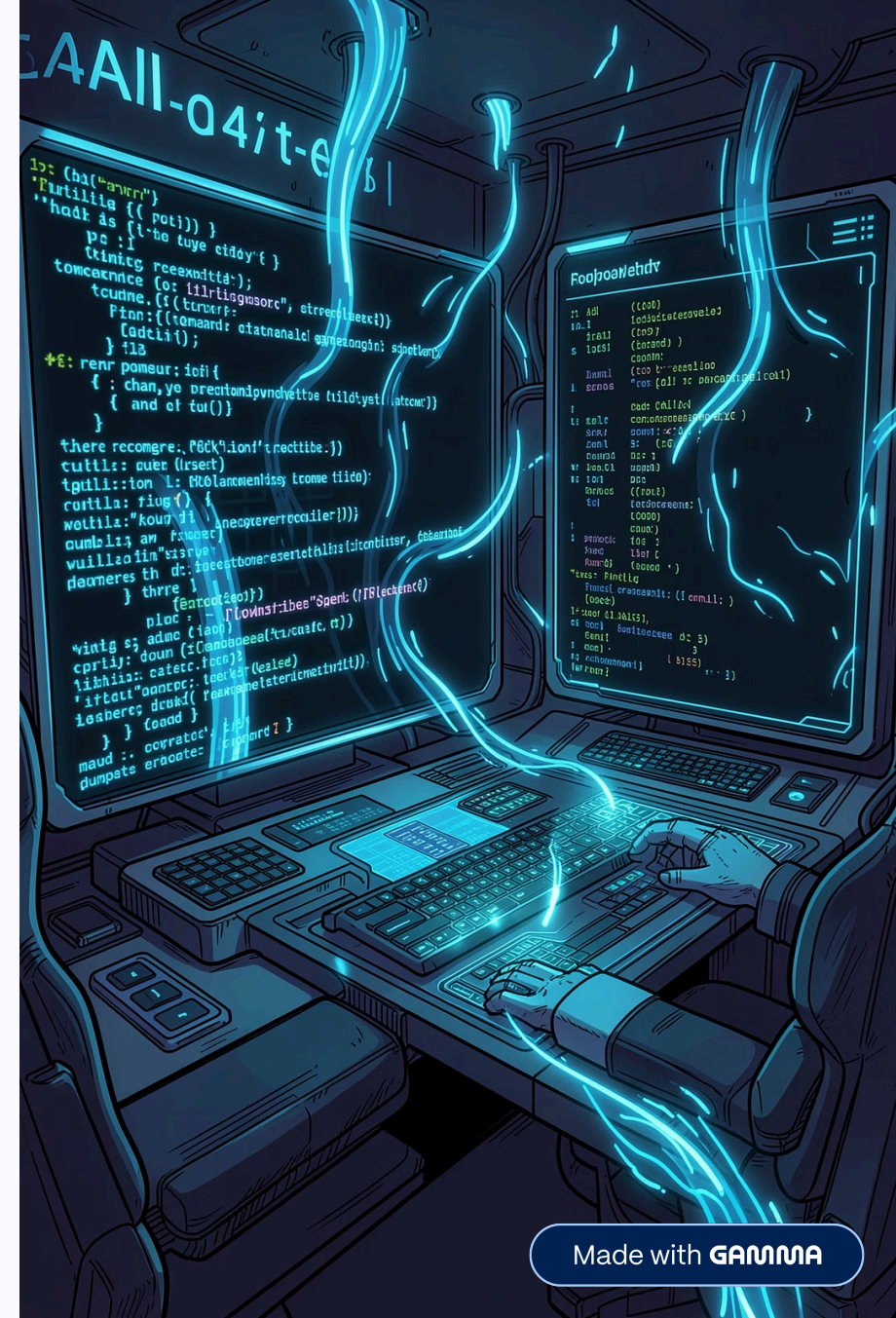
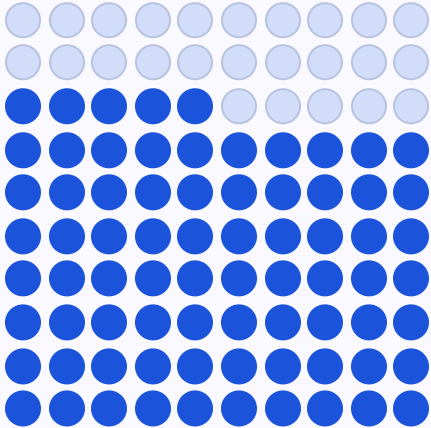


EMMA: Developer-Agent Interaction

How developers collaborate with EMMA – from goal input to autonomous code delivery – with safety gates at every step.



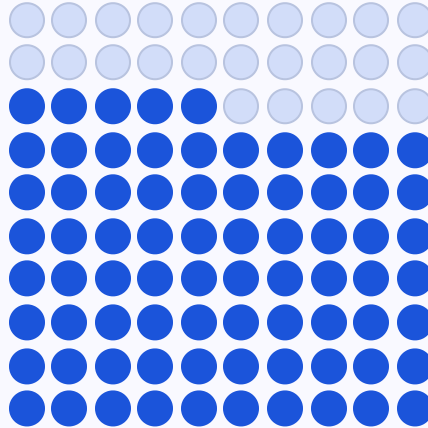
How Users Interact with EMMA



75

Goal Input

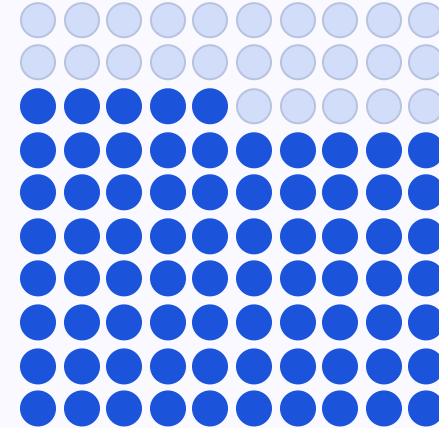
The developer inputs a task description, requirement, or JIRA ticket into the EMMA Cockpit.



75

Real-Time Cockpit Monitoring

The developer monitors the agent's step-by-step thinking, code mutations, and terminal output in real-time.



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Human-in-the-Loop (HITL) Gates

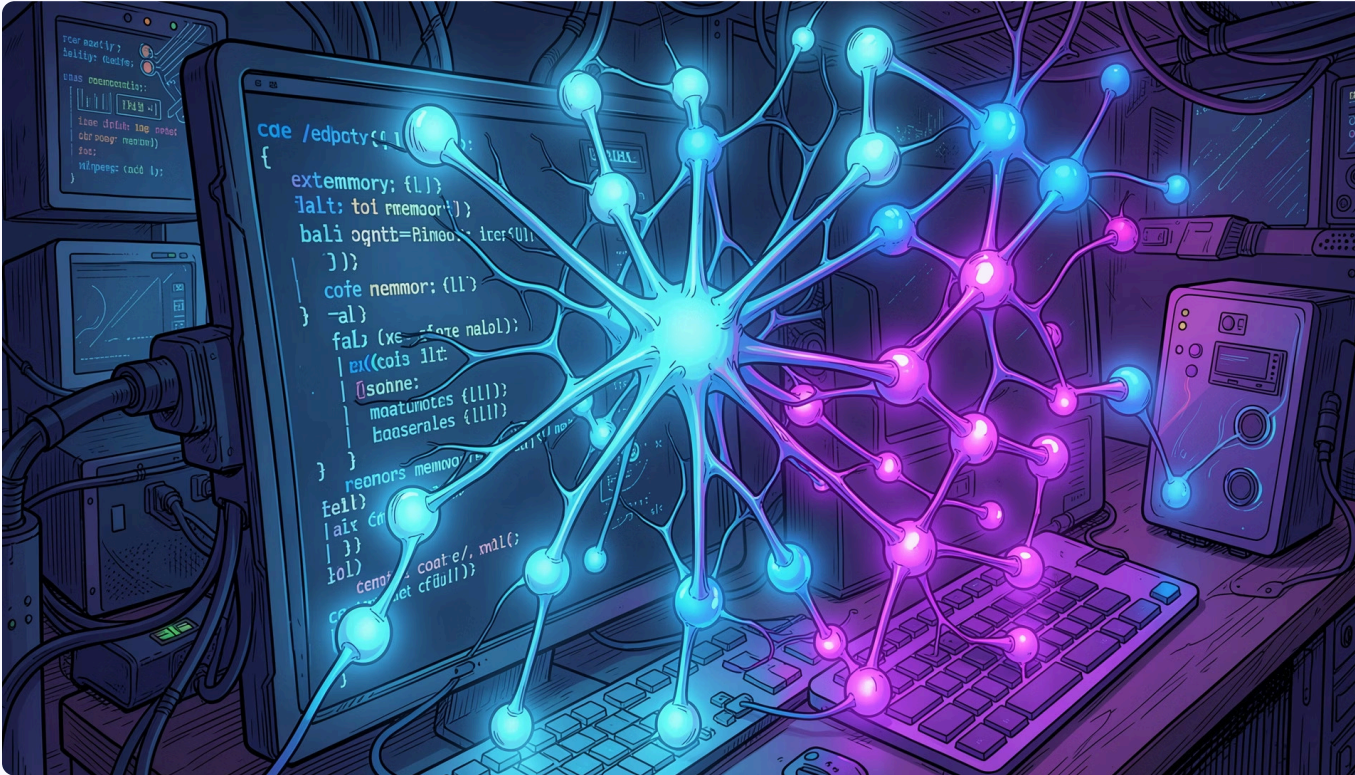
If the agent reaches a safety threshold or needs permission to modify critical files, it pauses and prompts the user for approval.

Five-Step Agent Pipeline



Each candidate passes through gated safety checks before any code touches the workspace.

Step 1: Ingest & Retrieve



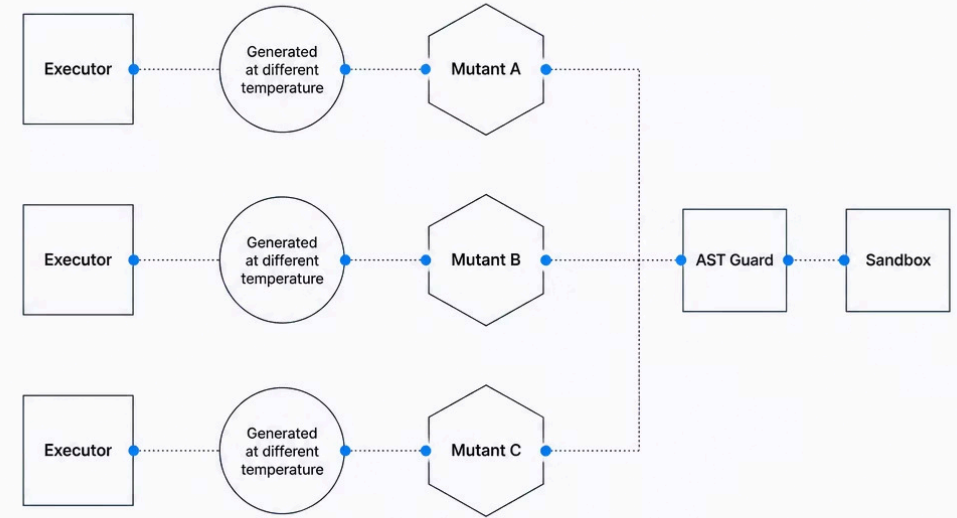
Orchestrator Solve Loop

The Orchestrator ingests the user request and queries the **LanceDB Memory Manifold** to retrieve similar historical code patterns – grounding each new task in prior learning.

Step 2: Parallel Mutation

Three Concurrent Proposals

The Executor spawns **three distinct code mutant proposals** simultaneously, each generated at a different temperature – enabling diverse solution strategies to compete in parallel.



Step 3: AST Guard & Sandbox Execution

Static Pre-Analysis

Code is pre-analyzed for complexity via AST parsing and Limits.py before any execution.

AST Gas Metering

Instrumented with Sudarshana AST Gas Counters to track computational cost and prevent runaway execution.

Jailed Subprocess

Executed inside an isolated subprocess sandbox – no host system access, no infinite loops, no OOM crashes.



Steps 4-5: Audit, Commit & Rollback



Adversarial Audit

The **Critic** evaluates terminal output, errors, and gas consumption to score each mutant.

Commit or Rollback

Pass: Candidate is committed to the workspace via **Git atomic commit**.

Fail: Workspace rolled back to last stable state; error cached to guide the next iteration.



Redrob Integration

Candidate Assessment Gate

EMMA's **AST Gas Metering Shield** integrates into Redrob's online programming challenge sandbox – safely running candidate submissions and preventing infinite loops or OOM hacks from crashing host servers.

Agent Benchmarking Workspace

Enables Redrob to offer next-generation **"AI Agent Assessments,"** running benchmark tasks against coding agents and ranking them for enterprise hiring.